

Image(s)

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Working Paper, 2024

Prepared for ifo Summer Course 2026, by Benjamin W. Arold

Motivation & Research Question

- Culture is not only attitudes and beliefs, but also 'a way of life' - how people dress, style their hair, and present themselves.
- Style choices signal status, group belonging, and individualism (Veblen 1899; Simmel 1957) - yet empirical work is rare for lack of high-frequency, granular data.
- Idea: use portrait photographs as a new data source to measure cultural change over time and space.
- Central questions: How similar are style choices across peers, across generations, and across places? When and where do influential new styles emerge?
- Method contribution: turn images into quantitative 'style vectors' and track similarity - individualism, persistence, and novelty.

Data: 14+ Million Yearbook Portraits

- Senior portraits of graduating US high-school students, 1930-2010, from ~112,000 yearbooks (classmates.com).
- About 14.5 million images across thousands of high schools in 44 states.
- A face/section-detection algorithm isolates senior portraits (99.6% correctly identified in human audits).
- Schools geocoded (OpenStreetMap) and aggregated to commuting zones; merged with census socioeconomics (ICPSR 2896) and patent data (Bell et al. 2019).
 - Not a random sample, but the near-universe of available senior images; regressions use commuting-zone fixed effects.

Method (1): From Portrait to Style Vector

Figure 1: Sample Images and Style Attributes from High School Yearbook

Style Feature	Model Dummies	
	Image A	Image B
long_hair	1	1
curtained_hair	1	0
curly_hair	0	0
spiky_hair	0	0
bowlcut_hair	0	0
crew-cut_hair	0	0
afro	0	0
beard	0	0
moustache	0	0
earrings	0	0
necklace	0	0
glasses	1	0
low_neckline	0	1
sweater	0	0
suit	0	0
shirt_collar	1	0
shirt_nocollar	0	0
bowtie	0	0
tie	0	0



Image A



Image B

Note: Two example portraits and their model-predicted style-attribute vectors. Each style element is a 0/1 dummy from the Vertex AI classifier. (Figure 1)

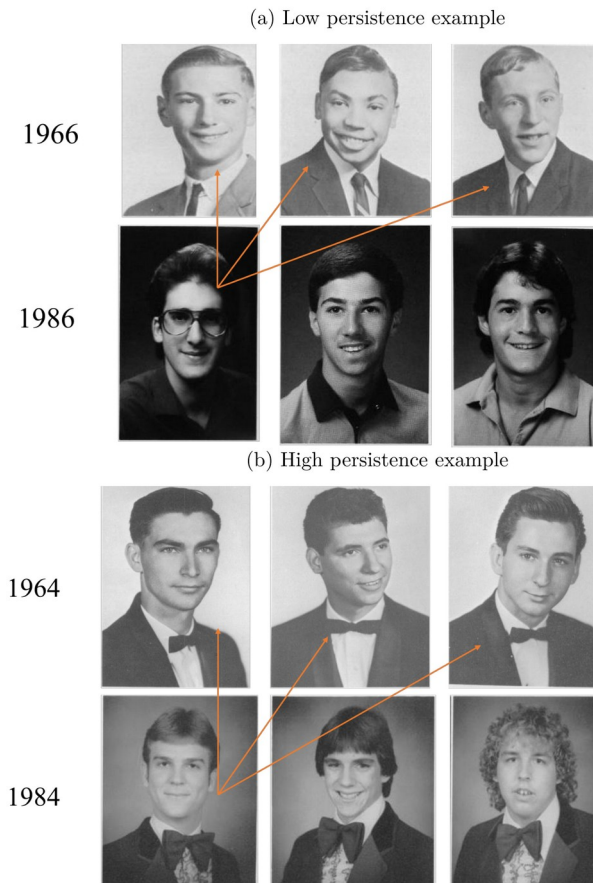
- Images converted to grayscale (to avoid color-film artifacts).
- A fixed vector of 25 style elements: hair length/style, facial hair, glasses, necklines, jewelry, ties, collars, suits.
- CNN multi-label classifiers (Google Vertex AI) trained on human (MTurk/Prolific) labels; scores >0.3 become 0/1 dummies.
- Out-of-sample accuracy ~79% (clothes) to 98% (facial hair).
- Result: a sparse, interpretable vector per person encoding observable style choices.

Method (2): Similarity -> Three Culture Measures

- Compare two people by the COSINE SIMILARITY of their style vectors (1 = identical, 0 = orthogonal).
- Individualism: 1 - average similarity to same-gender peers in the same school-year (low similarity = high individualism / less conformity).
- Persistence: average similarity to seniors at the SAME school 20 years earlier (high = intergenerational transmission).
- Style novelty: a style is 'novel' if fewer than 1% of all seniors ever chose it up to that year - captures innovation / 'daring to be different'.
- Validation: human-vs-algorithm agreement (66%) is statistically indistinguishable from human-vs-human agreement (73%) - passes a 'Turing test'.

Illustrating Persistence: Cosine Similarity across Generations

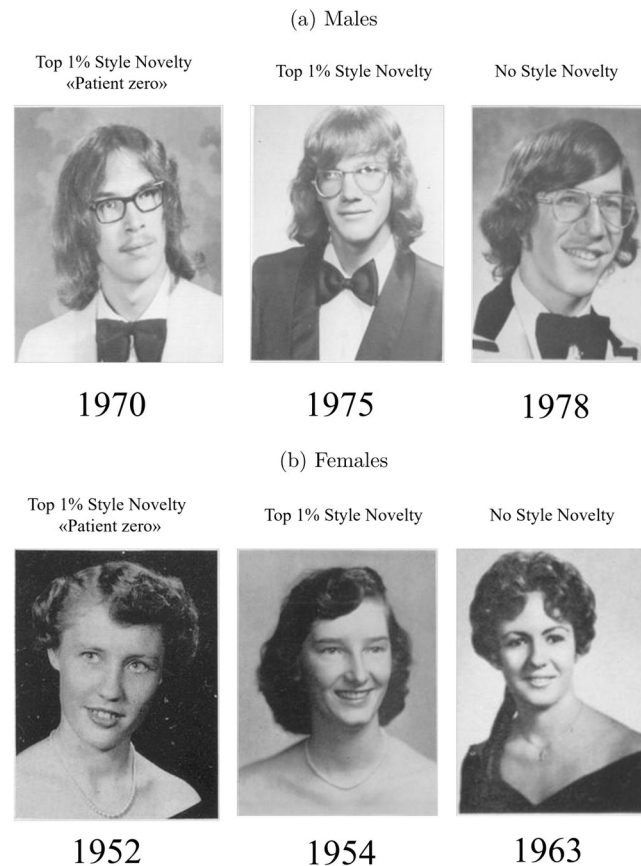
Figure 5: Sample Images and Persistence Calculation



Note: Persistence compares each senior with graduates of the same school 20 years earlier. Panel (a) East Providence HS: suits/ties give way to long hair and open shirts - persistence 0.056. Panel (b) De La Salle HS: bow ties and tuxedos endure across generations - persistence 0.83. (Figure 5)

Illustrating Style Novelty: Tracking a New Style's Diffusion

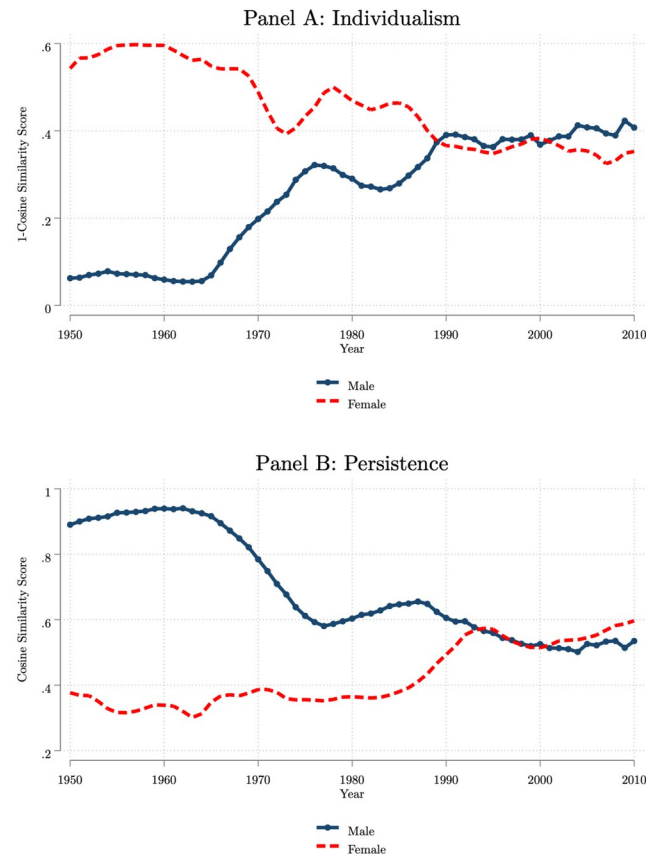
Figure 6: Sample Images and Style Novelty Example



Note: A style is novel while <1% of all prior seniors ever adopted it. Left = 'patient zero' (first appearance); as the style diffuses it stays top-1% (middle) until it becomes common and loses novelty (right). The method dates and locates influential style innovations. (Figure 6)

Result: Male-Female Convergence since the 1960s

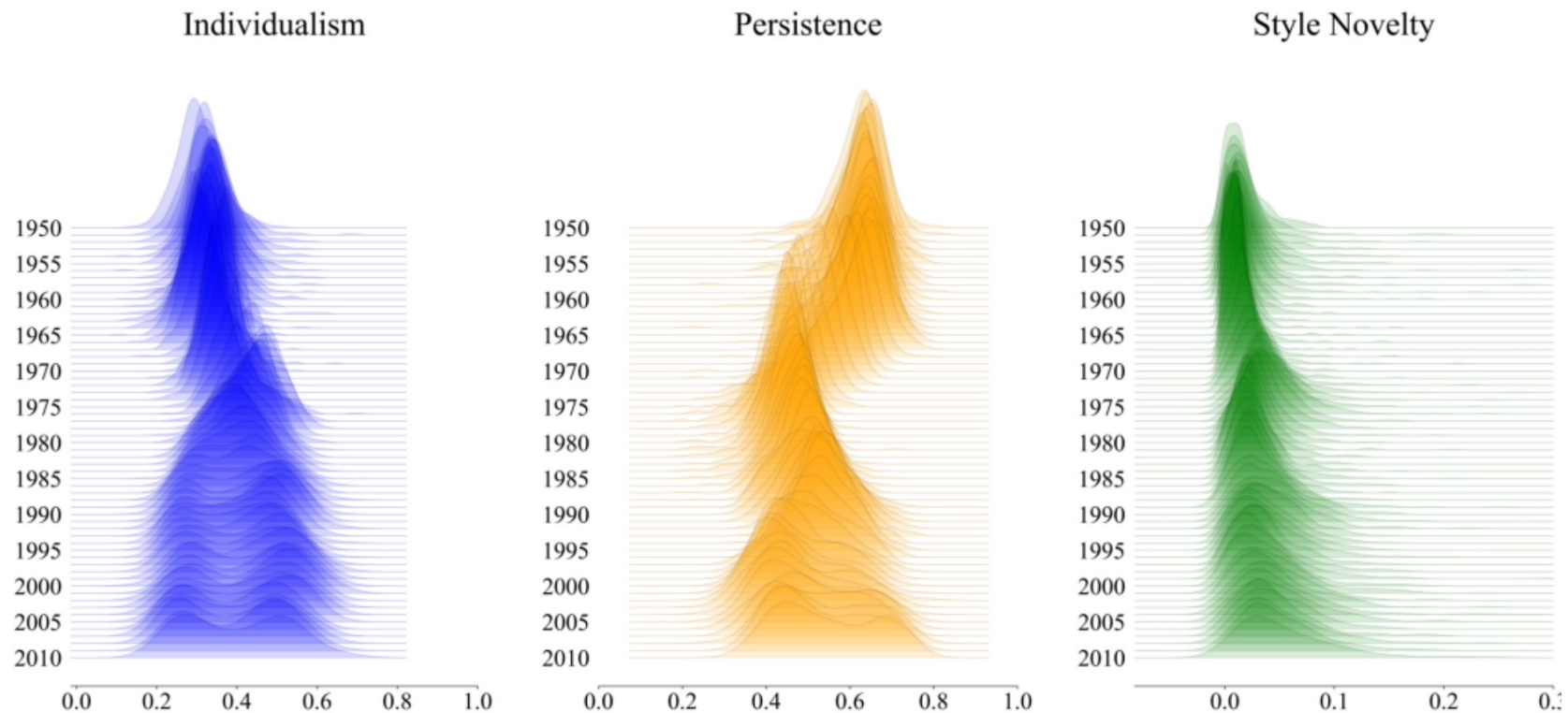
Figure 7: Individualism and Persistence Trends over Time, by Gender



Note: Panel A individualism, Panel B persistence, by gender. Men start conformist and highly persistent; from ~1965 (counterculture) male individualism surges and persistence collapses, converging with women by the 1990s. By 2010 the sexes are near-identical. (Figure 7)

Result: Rising Spatial Polarization across Commuting Zones

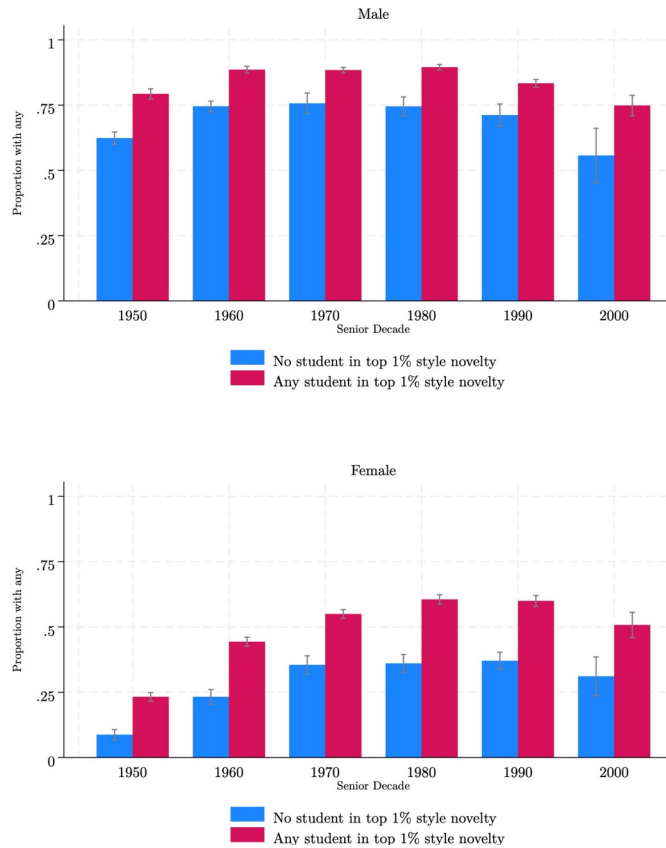
Figure 9: Variation across Commuting Zones



Note: Ridge plots of commuting-zone distributions by cohort. The uniform (unimodal) 1950s-60s give way to a stable BIMODAL split in individualism and persistence from the 1990s - some areas low-individualism/high-persistence, others the reverse; consistent with Gavrillets (2020). Style novelty disperses widely but is not bimodal. (Figure 9)

Result: Style Innovation Predicts Later Patenting

Figure 13: Patent Applications by Decade and Style Innovation



Note: Share of a commuting-zone cohort filing any patent, by senior decade, split by whether the cohort had a top-1% style innovator. Innovative areas patent more in every decade, for both men and women. A PCA index of style measures predicts ~ 0.16 SD more applications, robust to CZ and cohort fixed effects. (Figure 13)

Takeaways

- Images are a rich, high-frequency source for cultural economics: style vectors + cosine similarity yield tractable measures of culture.
- Three measures - individualism, persistence, style novelty - quantify conformity, intergenerational transmission, and innovation.
- Empirical facts: sharp male-female convergence, a collapse of male conformity around 1965-75, and rising spatial polarization since the 1970s.
- Lower local socioeconomic status predicts more cultural dynamism (higher individualism/novelty, lower persistence).
- 'Dress different' correlates with 'think different': style innovation predicts later patenting (not causal, but suggestive of a broader creative culture).